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DEPARTMENT OF INFORMATION TECHNOLOGY AND DECISION SCIENCES



AI POWERED FAULT DETECTION IN MINI – GRID RENEWABLE ENERGY

GROUP 24-B

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DEDICATION

This project is dedicated to Almighty God for His infinite grace and guidance,
and to our beloved parents for their unwavering support and encouragement throughout this
journey.

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ABSTRACT

The growing trend of renewable energy mini-grids in rural and underdeveloped regions has highlighted the need for fault detection systems that can guarantee uninterrupted power supply and longer lifespan of mini-grid renewable energy systems. Current fault detection technologies are largely based on manual inspections and rule-based systems. These technologies are highly prone to errors and are not suitable for real-time management of mini-grid renewable energy systems. This project proposes a solution to this problem by designing and developing an AI-based fault detection system that incorporates Machine Learning, Artificial Intelligence, and Internet of Things technologies. This fault detection system monitors and detects faults in mini-grid renewable energy systems. A simulated dataset is designed to mimic real-time mini-grid renewable energy system operational conditions. A Random Forest Machine Learning model is used to classify system states as normal or faulty. This project is implemented using Python and Scikit-Learn libraries. A web interface is designed using Flask, HTML, CSS, and JavaScript to display real-time data and instant fault detection. The implementation was carried out in a structured manner that involved backend development, machine learning model development and deployment, and interface development. Testing was carried out through unit testing, integration testing, system testing, and user acceptance testing. The results obtained indicate that the proposed system effectively classifies the states of the system with a high level of accuracy, thereby increasing the efficiency of fault detection in the system significantly. The proposed system provides a cost-effective solution to the enhancement of the reliability of renewable energy mini-grid systems, thereby contributing to the achievement of sustainable energy in the region.

Keywords: Fault Detection, Mini-Grid, Renewable Energy, Machine Learning, Random Forest, Internet of Things, Real-Time Monitoring

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LIST OF ABBREVIATIONS

AI	- Artificial Intelligence
API	- Application Programming Interface
CNN	- Convolutional Neural Network
CPU	- Central Processing Unit
CSV	- Comma-Separated Values
ESP32	- Espressif Systems 32-bit Microcontroller
Flask	- Python Web Framework
GUI	- Graphical User Interface
HTML	- Hypertext Markup Language
HTTP	- Hypertext Transfer Protocol
ICT	- Information and Communication Technology
IDE	- Integrated Development Environment
IoT	- Internet of Things
JSON	- JavaScript Object Notation
ML	- Machine Learning
PV	- Photovoltaic
RAM	- Random Access Memory
RF	- Random Forest
SCADA	- Supervisory Control and Data Acquisition
SDLC	- Software Development Life Cycle
SVM	- Support Vector Machine
UAT	- User Acceptance Testing
UI	- User Interface
URL	- Uniform Resource Locator

CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

The world is witnessing a revolution in the energy sector from fossil fuels to renewable energy sources in response to climate change and global warming, and to meet the increasing energy demands and reduce carbon emissions. Among all renewable energy sources, solar photovoltaic technology has gained more popularity and is growing rapidly, particularly in sunny regions.

In many developing countries like Ghana and other parts of sub-Saharan Africa, it is difficult and expensive to extend the national electricity grid to rural and remote areas. Renewable energy mini-grids are becoming a popular and promising solution to meet this need. A mini-grid is a small-scale electricity generation and distribution system that is often localized and operates independently or semi-independently of a public utility grid. In most mini-grids, solar energy is harnessed using photovoltaic systems combined with battery energy storage systems and inverters to supply power to households, schools, clinics, and small businesses.

Despite the socio-economic benefits that mini-grids offer, such as improved educational outcomes, enhanced healthcare facilities, and opportunities for small-scale enterprises, their sustainability over a long period of time is entirely dependent on their reliable functioning. Mini-grids are prone to several technical issues, including voltage stability, current imbalance, battery degradation, inverter failure, partial shading of solar cells, and overall system overheating, resulting from environmental factors like dust, high temperatures, and humidity, typical of tropical climates.

In many mini-grid systems, particularly in rural Ghana, fault detection is usually done by manually checking each system, although this is a very slow process. Some systems employ a more sophisticated fault detection technique, where a rule-based system is used to monitor parameters, sending a warning signal when a fault is detected by crossing a threshold. These techniques are usually inadequate, as small faults may develop into serious faults, causing prolonged outages, reduced equipment life, and higher maintenance costs. Operators of mini-grids in these areas may be unaware of how to interpret complex system information.

The development of Artificial Intelligence (AI) and Machine Learning (ML) presents new possibilities for addressing the challenges identified above. ML has the ability to learn the normal state of the system and recognize even the slightest abnormality that may be a precursor to a fault in the system. This presents a new opportunity for predictive and real-time fault detection in mini-grids. When combined with Internet of Things (IoT) technology and a user-friendly interface on the web, it presents a new opportunity for the improvement of mini-grid operation.

The purpose of this project was to contribute to the development of AI in the context of renewable mini-grids by developing a fault detection system that uses AI and ML to automatically identify the state of the mini-grid as normal or faulty.

1.2 Statement of the Problem

Currently, despite the rising trend in the installation of solar mini-grids in Ghana and other parts of sub-Saharan Africa, mini-grids are still marred by poor reliability because of inadequate fault detection systems. In traditional fault detection systems, fault detection is usually done when a fault occurs. In this system, faults are detected when they cause disruptions in power supply. This system is also quite expensive and difficult in areas where roads are poor. In this system, threshold detection is used. This system is also prone to giving false alarms because it is usually triggered when there is a fluctuation in power. This fluctuation could be caused by various factors such as cloud cover and battery capacity loss over time.

In addition to this, despite the availability of adequate information in terms of voltage, current, temperature, and battery state of charge, this information is not being used intelligently. In rural areas where mini-grids are usually located, this information is not being used because of a lack of tools and expertise.

In the Ghanaian context, studies have also pointed to challenges such as the lack of technical know-how among local technicians, challenges in the procurement of spare parts, as well as the effects of adverse environmental factors on the mini-grid systems. In the absence of intelligent monitoring systems, the sustainability of mini-grids is compromised, thereby slowing the pace towards universal access to electricity.

This project seeks to address the challenges by developing an intelligent system using AI, which will automate fault detection in real-time, thereby facilitating faster decision-making through the use of visualizations.

1.3 Aim and Objectives

Aim:

The aim of this project is to design and develop an AI-powered fault detection system for real-time monitoring and automated fault identification in renewable energy mini-grid systems.

Specific Objectives:

To examine the operational parameters of solar mini-grid systems and identify the key indicators related to common faults.

To develop a set of realistic simulated data and a machine learning model that has the ability to classify the state of the system as normal or faulty.

To develop a user-friendly web-based interface that has the ability to display the state of the system in real-time.

To develop a system that has the ability to integrate the backend and the frontend.

To examine the performance of the system in the context of accuracy, response time, and reliability.

1.4 Significance of the Study:

This project has a number of implications for different stakeholders in the mini-grid solar system. For instance, the mini-grid solar system has the ability to provide a tool that has the ability to reduce the cost of maintenance in the long run. In addition, the project has the ability to provide a reliable source of electricity for the local community, which has a number of positive implications for the local economy.

Academically, the project contributes to the increasing body of knowledge pertaining to the application of AI and ML in renewable energy systems in the context of developing countries. It also shows the potential of relatively accessible tools (Python, Scikit-learn, Flask) in addressing actual problems in the field of engineering. The project also has the potential to be used as a starting point for future academic studies that involve the actual integration of physical IoT devices, multi-classification of faults, or predictive maintenance.

1.5 Scope and Limitations

Scope:

The project scope involves the development of a software-based solution for fault detection in solar-based mini-grids, including dataset simulation, a binary random forest classifier, a web-based interface, and functional/performance-based testing.

Limitations:

The system has been developed and tested using simulated data rather than real-time field data of an operational mini-grid.

The dataset is of moderate size, and the results may not be generalized for highly varied environmental conditions.

The prototype does not include the integration of physical hardware (IoT sensors) in the system.

The classification of faults is only possible between normal and faulty conditions.

1.6 Organization of the Project

This document is organized into five chapters. Chapter One is for the background, problem statement, objectives, and scope of the project. Chapter Two is for the literature review of the concepts, theories, and identification of the gaps in the existing literature. Chapter Three is for the methodology employed in the development of the system. Chapter Four is for the implementation and results of the project (not included in this document), while Chapter Five is for the conclusions and recommendations.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

In this chapter, literature that is related to fault detection in renewable energy systems will be reviewed. It will include some of the major concepts and ideas that are associated with fault detection and some of the literature that has been written previously. It will be a combination of literature that has been written globally and some of the challenges that are experienced in mini-grids in SSA and Ghana.

2.2 Conceptual Framework

Some of the major concepts that will be used in this research include:

Renewable Energy Mini-Grids: A mini-grid is a small power system that is used for electricity generation and distribution. It is normally composed of solar PV panels and a battery and is mainly used in off-grid rural areas.

Fault Detection: It is a process that is used to identify any fault that is experienced in a system. Some of the faults that are experienced in mini-grids include line-to-line fault, ground fault, partial shading fault, and inverter fault.

Machine Learning for Classification: Supervised learning algorithms are applied here to learn patterns and then make classifications. In this case, the classification is between ‘Normal’ and ‘Faulty’ based on the sensor data.

Real-Time Monitoring and Dashboards: This is the process of continuously monitoring the system and collecting the data with minimal latency.

Internet of Things (IoT): IoT is the collection of sensors and other devices that collect data. Although this prototype is based on a simulation environment, in the future, IoT can be integrated with the system.

2.3 Theoretical Framework

Systems Theory

The mini-grid is considered a system of different components (solar panel, battery, inverter, load) that are all connected and can be affected if one of these components fails. For fault detection, it is essential to monitor all these components.

Pattern Recognition Theory

Pattern recognition theory is a basis for ML algorithms. In a normal state, a mini-grid will exhibit a certain pattern; conversely, a fault will create a different pattern or a change in the pattern.

Signal Processing Theory

The theory offers tools for pre-processing sensor readings (voltage, current, temperature) and transforming this data into a format that can be used by ML algorithms.

Decision Theory

The theory considers how information will be provided to a human user of the system; this information should be provided in a manner that is easy to understand and timely.

2.4 Review of Related Literature

Fault detection in PV systems has traditionally employed threshold-based techniques and visual inspections. These methods, although simple, are generally inadequate in dealing with changing conditions and degradation, which may cause delays in fault detection.

Recent research has focused on machine learning techniques for fault detection in PV systems. Various machine learning algorithms, including Support Vector Machines, K-Nearest Neighbors, Decision Trees, Random Forest, and Neural Networks, have been used for fault detection in PV systems. Among these, Random Forest has shown promising results for fault detection in PV systems. Its classification accuracy has been reported to be higher than 94–99% in different fault detection problems in PV systems due to its immunity to overtraining and ability to deal effectively with feature interactions.

For instance, a study on fault diagnosis of a PV array using Random Forest has reported high classification accuracy by analyzing voltage and current parameters. Similarly, different machine learning techniques, including ensemble and deep learning, have been used for fault detection in more complex conditions, including partial shading and inverter faults. IoT-based monitoring systems have been developed for PV systems, although they generally employ simple fault detection techniques.

In the context of Africa, mini-grid implementation also has its own set of challenges, like limited technical know-how, environmental conditions like dust and extreme temperatures, and issues with maintenance logistics. Another set of challenges relates to socio-technical issues like land, illegal connections, and lack of skilled technicians. Not much research has been done on the implementation of integrated AI technologies, especially those which are simple enough to be implemented in rural mini-grid environments.

Research on dashboards and visualizations highlights the significance of effective interfaces. Effective dashboards can reduce human cognition by using color codes, trend graphs, and alerts, thus helping in faster response times even for individuals with limited technical know-how.

2.5 Gap Analysis

While there has been significant work done in fault detection through ML techniques, there are still gaps in the research:

- ✓ **Integration Gap:** Most research has been done either on ML or on IoT monitoring; however, not much research has been done on providing a complete picture of an integrated intelligent fault detection system with a simple web-based interface for mini-grid operators.
- ✓ **Context Gap:** Most existing work focuses on utility-scale solar farms or large-scale grid-connected solar power systems. Limited work has been done to address the unique requirements and constraints of small-scale solar mini-grid power systems in rural developing countries like Ghana.
- ✓ **Usability and Cost Gap:** Advanced techniques might demand costly equipment or require highly skilled manpower, which might not be feasible in developing countries. Cost-effective prototypes with low implementation complexities are required, with a focus on simplicity rather than accuracy.
- ✓ **Data Realism Gap:** Many models are trained on unrealistic or limited data sets. Simulation-based techniques with realistic mini-grid conditions, including environmental conditions, are not yet explored in local environments.

CHAPTER THREE: METHODOLOGY

3.1 Introduction

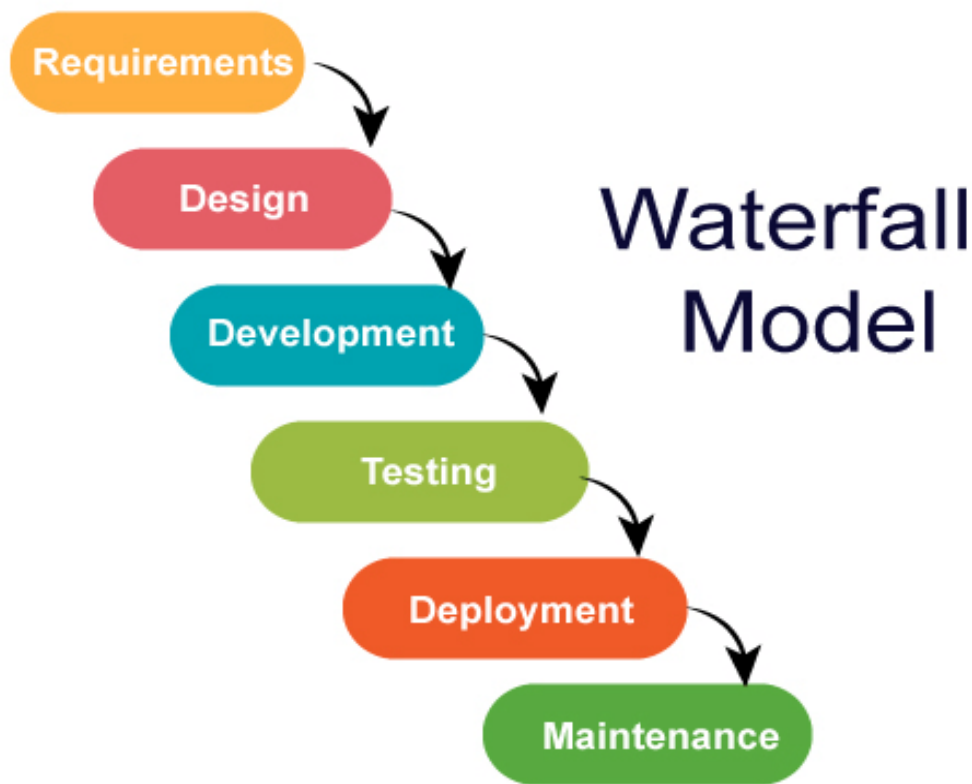
In this chapter, a description of the methodology adopted in developing the AI-powered fault detection system is provided. It includes a discussion of the software development methodology adopted, requirements gathering and analysis, system architecture, modeling diagrams, tools and technologies used, and finally the implementation plan. It is a systematic approach that ensures that the objectives are translated into a working prototype.

3.2 Software Development Methodology

The software development methodology adopted for this project is the Waterfall Model. It is a linear and sequential approach that consists of different stages or phases that are to be undertaken one after another.

This approach was appropriate because the requirements were well understood from the start, although they changed to a small extent. It also allowed for good documentation at each stage, which was required for the academic project in the final year. The stages implemented were:

Figure 3.1



Gathering and analysis of requirements
System design – architecture and modeling
Data preparation/model development
Implementation – backend and frontend development
Testing/evaluation
Documentation

These stages were implemented in sequence, with each stage producing intermediate results to be reviewed before moving to the next stage, thus reducing the chances of rework in the future.

3.3 Requirements Analysis

In-depth requirements analysis was done through literature review, consideration of the requirements from the mini-grid operators' perspective, and assessing the feasibility.

3.3.1 Functional Requirements

The system shall be able to:

Receive operational data from sensors such as voltage, current, temperature, and battery SOC through file upload or API.

Pre-process incoming data, classify faults, and generate output using a trained machine learning model.

Display real-time and historical data on a web-based interface.

Display visual and text-based alerts when a fault is classified by the machine learning model.

Provide a simple, intuitive interface that can be accessed by standard web browsers.

3.3.2 Non-Functional Requirements

Performance: The response time for classification by the machine learning model should be within 2 seconds.

- **Accuracy:** The machine learning model should be able to classify faults with at least a 90% accuracy rate.
- **Usability:** The interface should be simple and easy to understand for people with basic computer knowledge.
- **Scalability:** The system should be able to accommodate additional sensors or fault classes in the future.

- **Reliability:** The system should be able to operate continuously without frequent crashes.
- **Compatibility:** The system should be able to operate on a variety of operating systems and modern web browsers.

3.4 System Architecture and Design

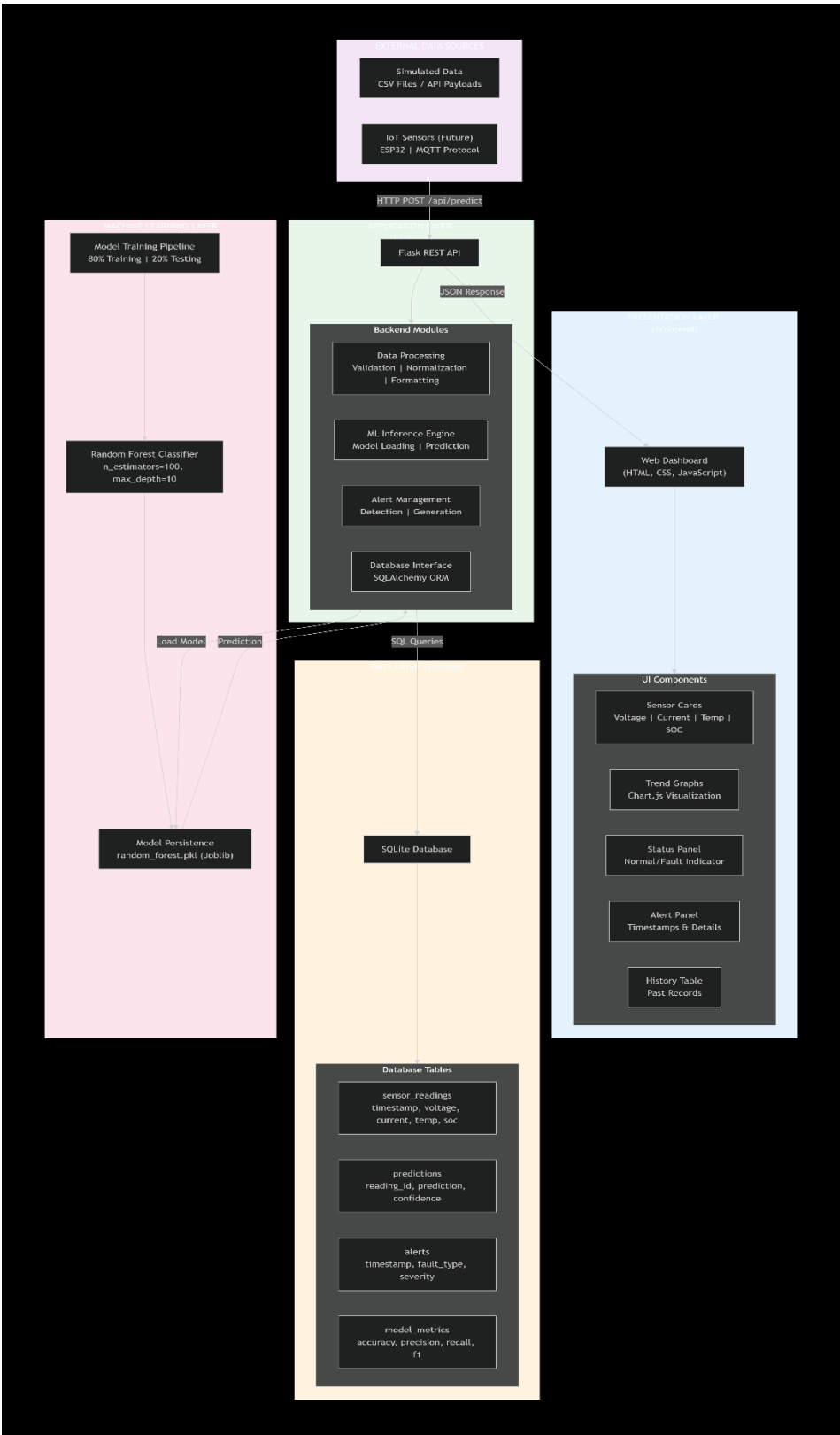
The system architecture will be a client-server architecture with a three-tier system, which provides a clean separation of concerns and makes the system easy to manage and scale:

Presentation Layer (Frontend): This refers to the web-based dashboard that provides the interface for the user to interact with the system, visualize the data (using charts and status indicators), and display alerts. This will be implemented using HTML, CSS, and JavaScript.

Application Layer (Backend): This refers to the logic that processes the input data, validates it, pre-processes it, makes predictions, and communicates with the API. This will be implemented using the Flask framework in Python, with the inclusion of the Random Forest model for predictions.

Data Layer: This refers to the storage and retrieval of historical data, which will be implemented using SQLite. Input data will also be accepted in the form of CSV or JSON for simulation purposes.

Figure 3.2



3.5 System Modeling Diagrams

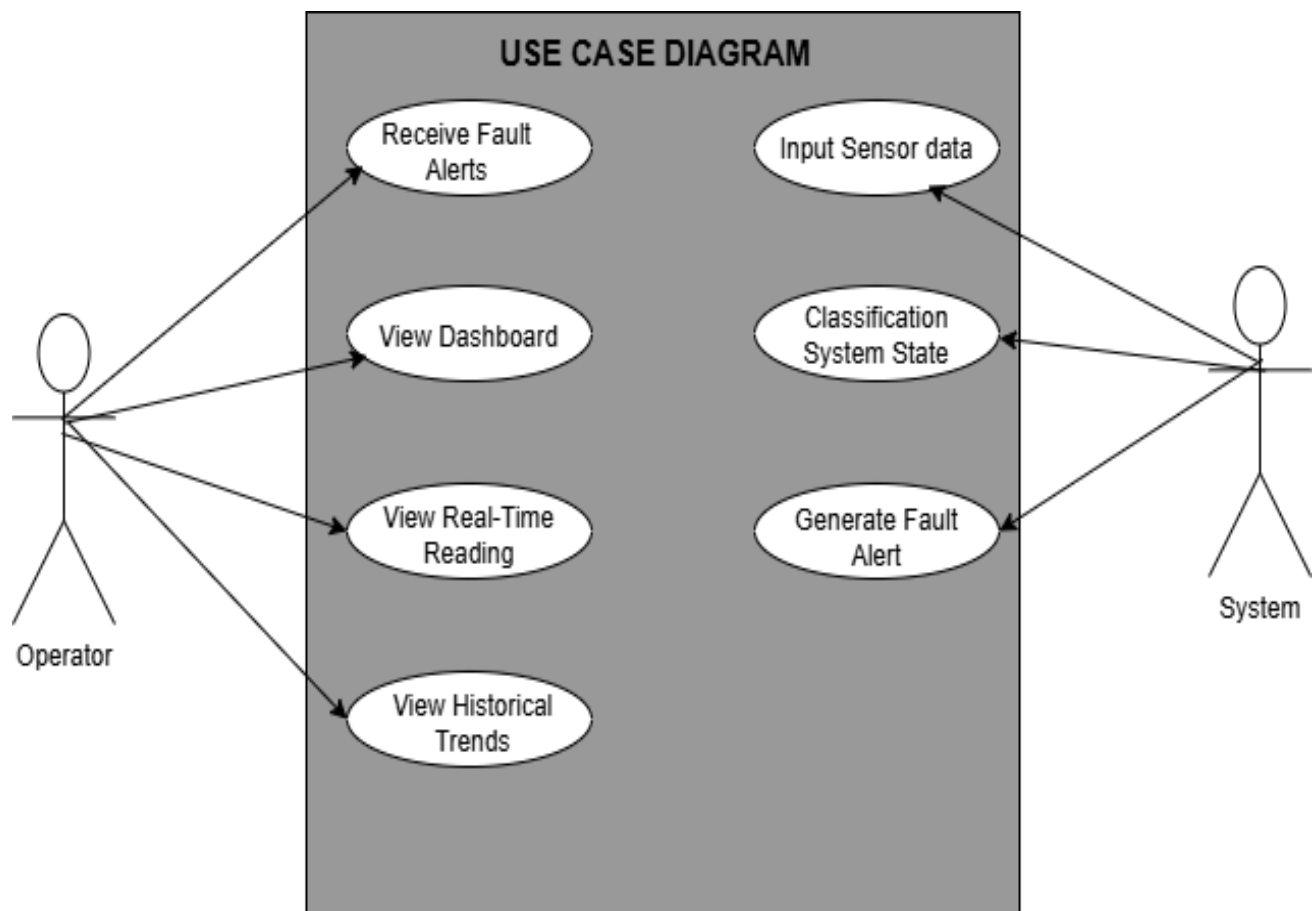
To clearly define system behavior and interactions, the following diagrams were developed:

3.5.1 Use Case Diagram

Actors include the Mini-Grid **Operator** and the **System** itself.

Key use cases are: Input Sensor Data, View Real-Time Dashboard, View Historical Trends, Receive Fault Alerts, and Classify System State.

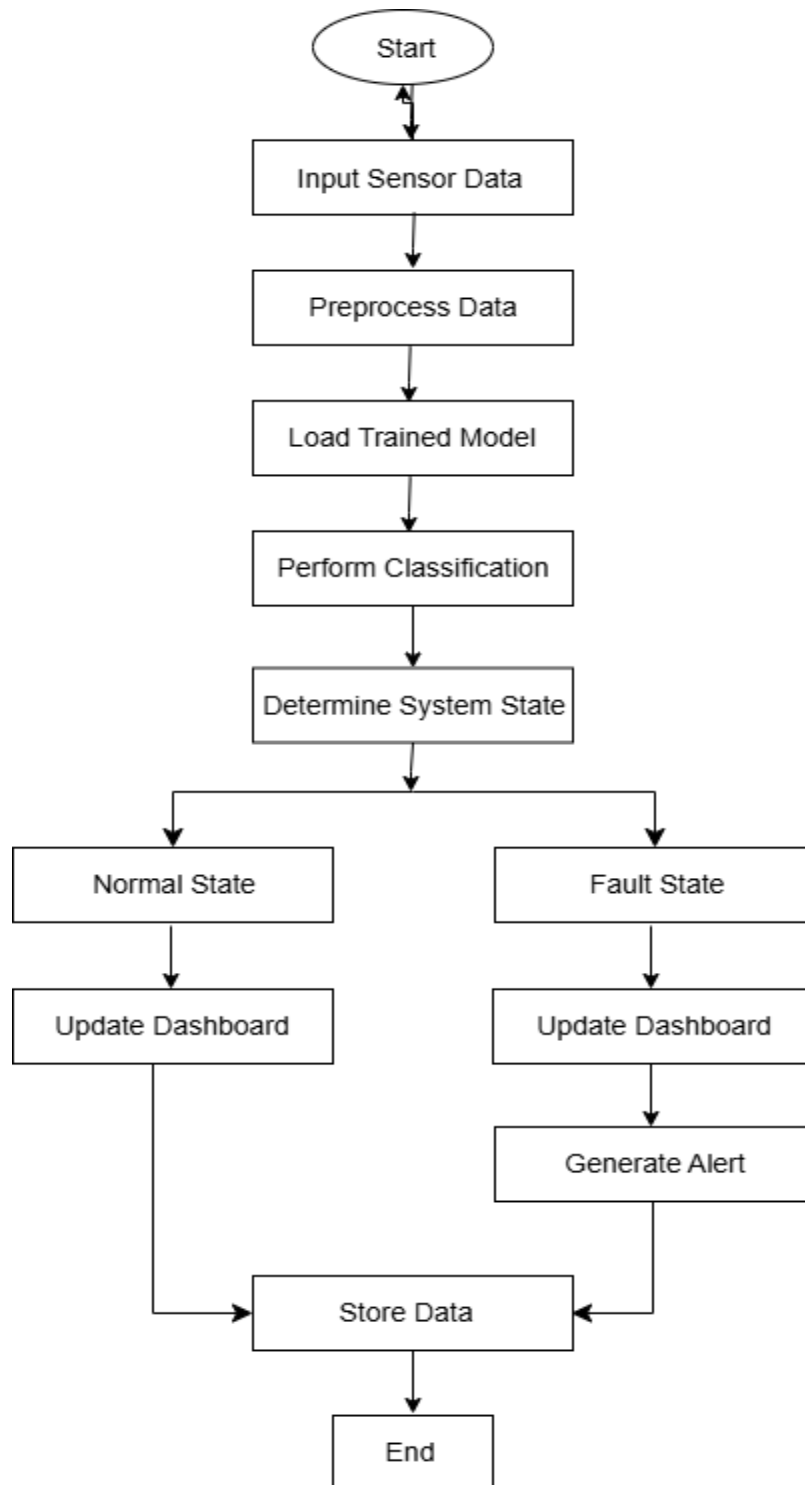
Figure 3.3



3.5.2 Activity Diagram for Fault Detection

The diagram below illustrates the step-by-step flow: data input → preprocessing → model loading → classification → dashboard update → alert generation (if faulty) → data storage.

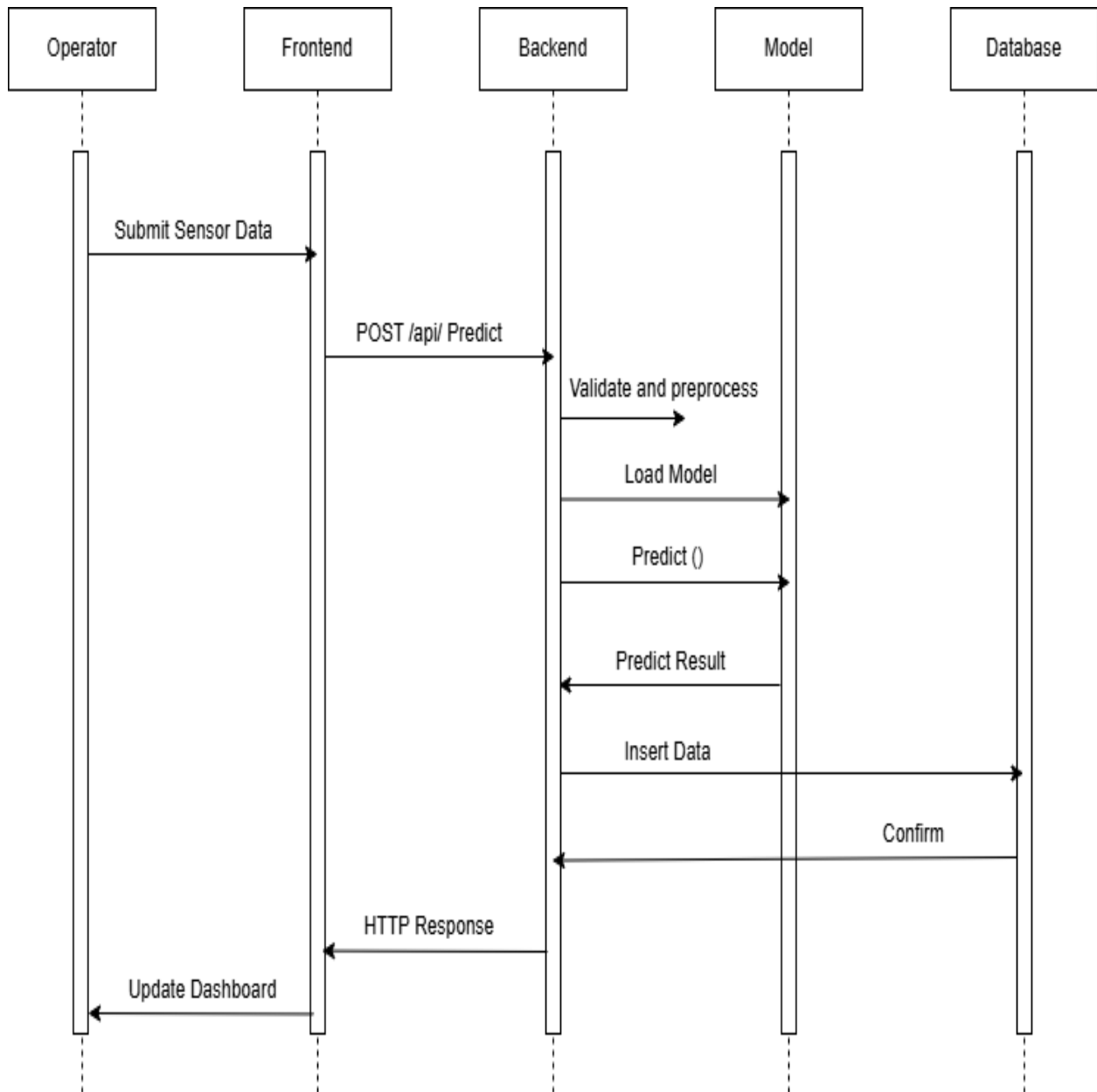
Figure 3.4



3.5.3 Sequence Diagram for Data Processing

Shows interactions between the Frontend, Backend API, Machine Learning Model, and Database when a user submits data or the system processes new readings.

Figure 3.5



3.6 Tools and Technologies

The tools and technologies selected are accessible, efficient, and appropriate for a final year project.

Programming Language: Python 3.9+

The language has a strong library support for data science and web development.

Web Framework: Flask

Flask is a lightweight and flexible framework for developing RESTful APIs.

Machine Learning: Scikit-learn

The library is used for Random Forest implementation, training, and evaluation.

Data Handling: Pandas and NumPy

These libraries are used for data creation and preprocessing.

Model Persistence: Joblib

The library is used for saving and loading the trained Random Forest model.

Frontend: HTML5, CSS3, JavaScript, and Chart.js

These are used for frontend development and creating interactive charts and visualizations.

Database: SQLite

The database is used for storing and managing data. It is a lightweight and serverless database.

Development Environment: Visual Studio Code and Git

The code editor is used for coding and version control using Git.

3.7 Implementation Plan

The project was carried out in a series of structured phases over a period of weeks as follows:

Phase 1: Requirements analysis and literature survey

Phase 2: System design and modeling diagrams

Phase 3: Creation of datasets and data preprocessing

Phase 4: Development of the machine learning models

Phase 5: Development of the backend

Phase 6: Development of the frontend dashboard

Phase 7: Testing phase

Phase 8: Documentation phase

3.8 Ethical Considerations

In spite of the fact that the project is more simulation-oriented, ethical issues have been taken into account. In this case, the use of simulated data is applied in order to avoid issues of privacy with the operational mini-grid data. In addition, the system is only for monitoring and alerting purposes and does not control any equipment in a physical manner in order to maintain the importance of human oversight and intervention in safety-critical operations. It is also made clear that the alerts are generated by the AI.